📘 Node.js Beginner Guide

1 What is Node.js?

Node.js is a JavaScript runtime that allows you to run JavaScript on the server (backend).

Used for:
- APIs
- Backend services
- Database operations
- Authentication

2 Install Node.js

Step 1: Download
Go to 👉 https://nodejs.org
Download LTS version

Step 2: Install
Click Next → Next → Finish
No need to change settings

Step 3: Check Installation
Open Command Prompt / Terminal
node -v
npm -v

✓ If versions appear → Node.js installed successfully

3 Create Node.js Project

Step 1: Create Project Folder
mkdir my-node-app
cd my-node-app

Step 2: Initialize Project
npm init -y
This creates:
- package.json

**What is PATH & Why It's Important?

PATH allows you to run node and npm commands from anywhere in the system.

** **How to Check PATH (Windows)**

🔷 **Step 1:**

- Press **Win + R**
- Type **sysdm.cpl**
- Enter

🔷 **Step 2:**

- Advanced tab → **Environment Variables**
- Select **Path**
- Click **Edit**

🔷 **Step 3:**

Check this path exists:

C:\Program Files\nodejs\

If not → Add it → Save → Restart PC

4️⃣ Install Required Packages

npm install express dotenv cors

npm install nodemon --save-dev

Packages Meaning

Package	Use
express	Web framework
dotenv	Environment variables
cors	Allow frontend access
nodemon	Auto server restart

5 Folder Structure (As You Requested)

```
my-node-app
│
├── controller
│   └── userController.js
│
├── router
│   └── userRouter.js
│
├── index.js
├── app.js
├── server.js
│
├── .env
├── package.json
```

6 Path Setup & File Responsibilities

File	Purpose
server.js	Start server
app.js	App configuration
index.js	Route connection
router	API routes
controller	Business logic

1. controller/userController.js

```
const getUsers = (req, res) => {
  res.json({
    message: "User API Working",
  });
};

module.exports = {
  getUsers,
};
```

2.router/userRouter.js

```
const express = require("express");
const router = express.Router();

const {
  getUsers,
} = require("../controller/userController");

router.get("/getUsers", getUsers);

module.exports = router;
```

3.index.js (Main Router)

```
const express = require("express");
const router = express.Router();

const userRoutes = require("./router/userRouter");

router.use("/users", userRoutes);

module.exports = router;
```

4.app.js (App Setup)

```
const express = require("express");
const cors = require("cors");
require("dotenv").config();

const indexRoutes = require("./index");

const app = express();

app.use(cors());
app.use(express.json());

app.use("/api", indexRoutes);
```

```
module.exports = app;
```

5. server.js (Server Start)

```
const app = require("./app");

const PORT = process.env.PORT || 3000;

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

1️⃣2️⃣ Database Connection (MongoDB Example)

Install MongoDB Package
npm install mongoose

****Create Database File**
📁 config/db.js (optional but recommended)

```
const mongoose = require("mongoose");

const connectDB = async () => {
  try {
    await mongoose.connect(process.env.MONGO_URI);
    console.log("MongoDB Connected");
  } catch (error) {
    console.error("DB Error:", error.message);
    process.exit(1);
  }
};

module.exports = connectDB;
```

1️⃣3️⃣ .env File

```
PORT=3000
MONGO_URI=mongodb://localhost:27017/mydb
```

1️⃣4️⃣ Run the Project

Add Script in package.json

```
"scripts": {
  "start": "node server.js",
  "dev": "nodemon server.js"
}
```

Start Server

```
npm run dev
```

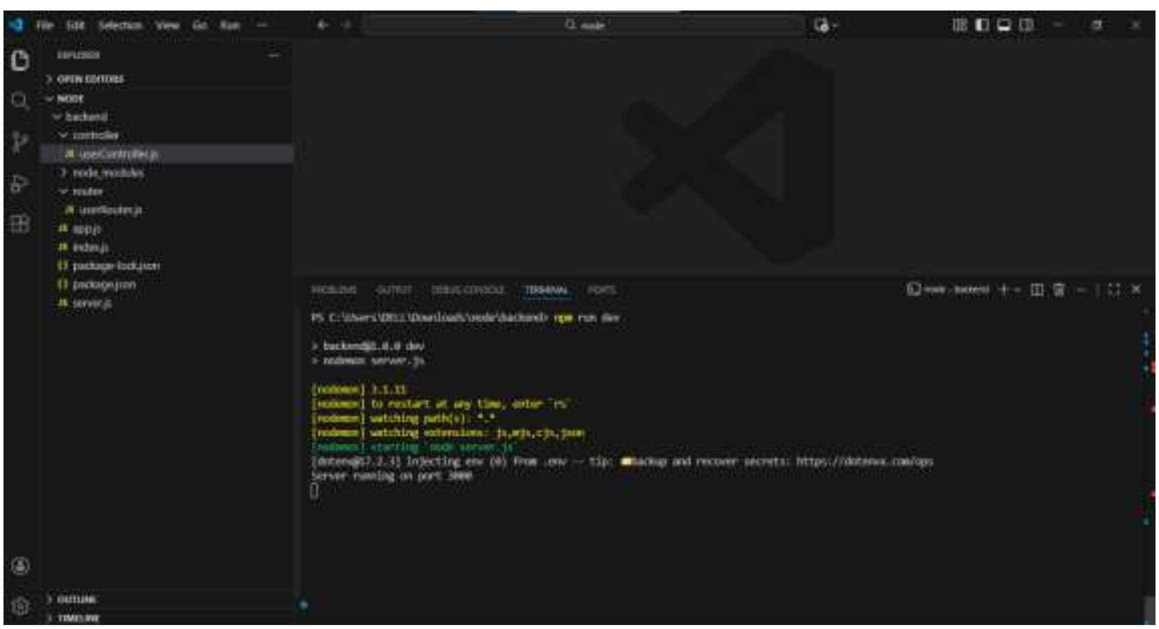
1️⃣5️⃣ Test API in Browser

Open:
http://localhost:3000/api/users

Output:

```
{
  "message": "User API Working"
}
```

Files Structures in Visual Studio



Test API in Postman